

Biology · Medium paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

120 marks

reading 15 min · writing 150 min

State average on these questions

75 / 120 (62%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the middle band of questions.

Sources

- [2022 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2023 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2024 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2025 Biology exam \(PDF\)](#) · [examiners' report](#)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗

Length of the translated molecule from exon and intron lengths

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 9 · [open in the official exam PDF \(p. 7\)](#) ↗

Location of the photosynthetic stage shown in a diagram

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 11 · [open in the official exam PDF \(p. 8\)](#) ↗

What raises the maximum rate of photosynthesis on a light-response curve

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 14 · [open in the official exam PDF \(p. 10\)](#) ↗

Naming compounds at labelled points of a respiration diagram

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 5 · 1 mark

2022 VCAA exam · Section A · Question 15 · [open in the official exam PDF \(p. 11\)](#) ↗

The loaded carrier molecules in a Krebs cycle diagram

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 6 · 1 mark

2022 VCAA exam · Section A · Question 19 · [open in the official exam PDF \(p. 12\)](#) ↗

A true statement about the enzymes of respiration and photosynthesis

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 7 · 1 mark

2022 VCAA exam · Section A · Question 28 · [open in the official exam PDF \(p. 15\)](#) ↗

A practical way to stop Hendra virus reaching horses from bats

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2022 VCAA exam · Section A · Question 31 · [open in the official exam PDF \(p. 16\)](#) ↗

Bioethical principle behind reporting all results

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark

2022 VCAA exam · Section A · Question 36 · [open in the official exam PDF \(p. 19\)](#) ↗

Describing the same bone arrangement in different vertebrate forelimbs

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark

2022 VCAA exam · Section A · Question 38 · [open in the official exam PDF \(p. 20\)](#) ↗

Comparing two differently drawn vertebrate phylogenies

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark

2022 VCAA exam · Section A · Question 40 · [open in the official exam PDF \(p. 21\)](#) ↗

Why mtDNA is useful for tracing migration routes

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark

2023 VCAA exam · Section A · Question 2 · [open in the official exam PDF \(p. 2\)](#) ↗

Interpreting labelled diagrams of two nucleotides

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark

2023 VCAA exam · Section A · Question 5 · [open in the official exam PDF \(p. 4\)](#) ↗

Roles of the organelles in exocytosis of a protein

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark

2023 VCAA exam · Section A · Question 6 · [open in the official exam PDF \(p. 4\)](#) ↗

What students should expect about proteins across different cell types

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark

2023 VCAA exam · Section A · Question 7 · [open in the official exam PDF \(p. 4\)](#) ↗

What systematic errors do

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark

2023 VCAA exam · Section A · Question 8 · [open in the official exam PDF \(p. 5\)](#) ↗

The most likely entry route for a fungal pathogen into a leaf

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark

2023 VCAA exam · Section A · Question 10 · [open in the official exam PDF \(p. 5\)](#) ↗

What gel electrophoresis does to DNA fragments

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark

2023 VCAA exam · Section A · Question 15 · [open in the official exam PDF \(p. 7\)](#) ↗

Molecules involved in the light-independent reactions

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark

2023 VCAA exam · Section A · Question 18 · [open in the official exam PDF \(p. 8\)](#) ↗

Conditions that give the highest rate of lactate formation

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark

2023 VCAA exam · Section A · Question 23 · [open in the official exam PDF \(p. 9\)](#) ↗

What an anti-inflammatory like tea-tree oil inhibits

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 21 · 1 mark

2023 VCAA exam · Section A · Question 25 · [open in the official exam PDF \(p. 10\)](#) ↗

What can be concluded about the control group in an allergy-treatment study

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 22 · 1 mark

2023 VCAA exam · Section A · Question 27 · [open in the official exam PDF \(p. 12\)](#) ↗

Advantage of humanising a mouse monoclonal antibody

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 23 · 1 mark

2023 VCAA exam · Section A · Question 29 · [open in the official exam PDF \(p. 12\)](#) ↗

An event that can occur in the lymphatic system

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 24 · 1 mark

2023 VCAA exam · Section A · Question 31 · [open in the official exam PDF \(p. 13\)](#) ↗

Type of immunity from a toxoid (inactivated toxin) vaccine

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 25 · 1 mark

2023 VCAA exam · Section A · Question 34 · [open in the official exam PDF \(p. 15\)](#) ↗

The least effective measure against mosquito-borne Zika

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 26 · 1 mark

2023 VCAA exam · Section A · Question 36 · [open in the official exam PDF \(p. 15\)](#) ↗

The least useful evidence for building a phylogeny of mammals

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 27 · 1 mark

2023 VCAA exam · Section A · Question 38 · [open in the official exam PDF \(p. 17\)](#) ↗

Nested circles showing mammals, primates, hominoids and hominins

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 28 · 1 mark

2023 VCAA exam · Section A · Question 39 · [open in the official exam PDF \(p. 18\)](#) ↗

Picking Homo sapiens on a log brain mass–body mass graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 29 · 1 mark

2023 VCAA exam · Section A · Question 40 · [open in the official exam PDF \(p. 19\)](#) ↗

Interpreting carrying angles of the femur in humans, australopithecines and chimps

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 30 · 1 mark

2024 VCAA exam · Section A · Question 11 · [open in the official exam PDF \(p. 7\)](#) ↗

Outputs of the electron transport chain and the role of NADH

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 31 · 1 mark

2024 VCAA exam · Section A · Question 14 · [open in the official exam PDF \(p. 8\)](#) ↗

Where water is split using absorbed light energy

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 32 · 1 mark

2024 VCAA exam · Section A · Question 16 · [open in the official exam PDF \(p. 9\)](#) ↗

A true statement about bioethanol production from biomass

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 33 · 1 mark

2024 VCAA exam · Section A · Question 22 · [open in the official exam PDF \(p. 12\)](#) ↗

Matching a type of immunity to how it was gained

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 34 · 1 mark

2024 VCAA exam · Section A · Question 25 · [open in the official exam PDF \(p. 13\)](#) ↗

Identifying the cell that binds an antigen on an APC and then activates another cell

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 35 · 1 mark

2024 VCAA exam · Section A · Question 30 · [open in the official exam PDF \(p. 15\)](#) ↗

Why tibia length changed across generations when scientists chose which mice bred

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 36 · 1 mark

2025 VCAA exam · Section A · Question 10 · [open in the official exam PDF \(p. 8\)](#) ↗

The other output of fermentation for bioethanol and the conditions needed

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 37 · 1 mark

2025 VCAA exam · Section A · Question 13 · [open in the official exam PDF \(p. 9\)](#) ↗

A true statement about PCR

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 38 · 1 mark

2025 VCAA exam · Section A · Question 18 · [open in the official exam PDF \(p. 11\)](#) ↗

A true statement about the lactic-acid fermentation stage

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 39 · 1 mark

2025 VCAA exam · Section A · Question 35 · [open in the official exam PDF \(p. 19\)](#) ↗

Which visible feature classifies a lemur as a primate

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 40 · 1 mark

2025 VCAA exam · Section A · Question 36 · [open in the official exam PDF \(p. 19\)](#) ↗

What a fossil with both reptile and mammal features can be used for

Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Short-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (7 marks)

2022 VCAA exam · Section B · Question 4 · [open in the official exam PDF \(p. 28\)](#) ↗

Exporting ovalbumin from chicken cells; ethics of intensive farming vs recombinant protein

a. Roles of organelles in exporting ovalbumin (3 marks)

b. Discuss an ethical concern, propose a solution and name the ethical concept (4 marks)

Question 2 (7 marks)

2022 VCAA exam · Section B · Question 6 · [open in the official exam PDF \(p. 30\)](#) ↗

Influenza: natural killer vs cytotoxic T cells; antigenic drift and yearly vaccination

a. Compare the roles of natural killer cells and cytotoxic T cells (4 marks)

b.i. What change in the virus alters the antigen (1 mark)

b.ii. Why flu vaccination is yearly, in terms of antigenic drift (2 marks)

Question 3 (11 marks)

2023 VCAA exam · Section B · Question 6 · [open in the official exam PDF \(p. 30\)](#) ↗

Making human insulin with recombinant plasmids; comparing insulin activity profiles; protein structure

a. Outline how human insulin is made using recombinant plasmids (5 marks)

b. Which insulin is rapid-acting compared with regular, with graph data (3 marks)

c.i. Hierarchical level of insulin (1 mark)

c.ii. What to consider when changing insulin's primary structure, with justification (2 marks)

Question 4 (5 marks)

2023 VCAA exam · Section B · Question 9 · [open in the official exam PDF \(p. 34\)](#) ↗

Two Lord Howe Island palm species: flowering times, soil pH and altitude as drivers of speciation

a. Two factors in the flowering data that may have contributed to speciation (2 marks)

b.i. Ideal pH and altitude for each species from the graphs (1 mark)

b.ii. How the pH and altitude differences could have led to speciation (2 marks)

Question 5 (7 marks)

2023 VCAA exam · Section B · Question 11 · [open in the official exam PDF \(p. 39\)](#) ↗

Lenski's long-term *E. coli* evolution experiment: controls, a citrate-using mutant, designing a follow-up

a. Two controlled factors (2 marks)

b. Expected effect of the citrate-using mutation on later generations, and why (2 marks)

c. Design an experiment on glucose concentration and the frequency of the citrate phenotype (3 marks)

Question 6 (4 marks)

2024 VCAA exam · Section B · Question 6 · [open in the official exam PDF \(p. 33\)](#) ↗

Two monoclonal antibodies binding different sites on the HER2 receptor of breast cancer cells

a. Compare the variable regions of the two antibodies (2 marks)

b. Two ways the mAbs could help treat the cancer (2 marks)

Question 7 (10 marks)

2024 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 34\)](#) ↗

Measles in Australia: why cases fell, a 2024 outbreak alert, and factors in emerging pathogens

a. An action that may explain the fall in cases from 1994, and how (3 marks)

b. A likely source of the new cases and two benefits of listing locations (3 marks)

c. Describe two factors driving the emergence of new pathogens (4 marks)

Question 8 (5 marks)

2025 VCAA exam · Section B · Question 1 · [open in the official exam PDF \(p. 24\)](#) ↗

mRNA processing of the insulin gene and steps in recombinant insulin production

a. Draw the processed mRNA and label three modifications (3 marks)

b. Complete steps 4 and 8 of recombinant insulin production (2 marks)

Question 9 (10 marks)

2025 VCAA exam · Section B · Question 6 · [open in the official exam PDF \(p. 32\)](#) ↗

RSV vaccination in pregnancy: type of infant immunity, boosters, economics, monoclonal antibodies, reducing spread

a. Type of immunity the babies get, justified (2 marks)

b. Why a booster is advised in a later pregnancy (2 marks)

c. One economic benefit of vaccinating pregnant women (2 marks)

d. How monoclonal antibodies reduce RSV severity in infants (2 marks)

e. Another way to reduce RSV spread to infants, justified (2 marks)

Question 10 (14 marks)

2025 VCAA exam · Section B · Question 8 · [open in the official exam PDF \(p. 36\)](#) ↗

Article: cyanobacteria injected into tadpoles restart brain activity in the light, covering photosynthesis, respiration and ethics

a. Two reasons tadpoles were chosen (2 marks)

b.i. Two molecules other than enzymes that cyanobacteria need to produce oxygen (2 marks)

b.ii. Role of each molecule in producing oxygen (2 marks)

c. Why brain activity stopped under long hypoxia (2 marks)

d. Why brain activity took time to restart, using a biochemical pathway (2 marks)

e. Two events along the electron transport chain (2 marks)

f. Which student's ethical view (beneficence vs non-maleficence) you agree with, justified from the article (2 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	A	Only the exons stay in mature mRNA: $50 + 70 + 120 = 240$ bases.	77% correct A 77% · B 10% · C 5% · D 9%
☐	2	B	The stage shown (light-dependent) happens on the thylakoid membranes.	75% correct A 5% · B 75% · C 12% · D 8%
☐	3	B	At the plateau light is no longer limiting. More chloroplasts (more photosynthetic machinery) raises the maximum. Filters or distance reduce light, and cooling slows the enzymes. <i>Most popular wrong answer: A (16%).</i>	74% correct A 16% · B 74% · C 3% · D 7%
☐	4	C	Glycolysis makes pyruvate, the Krebs cycle releases CO_2 , and the ETC produces water. <i>Most popular wrong answer: D (15%).</i>	72% correct A 6% · B 7% · C 72% · D 15%
☐	5	C	The Krebs cycle produces the loaded carriers NADH and FADH_2 , which take electrons to the ETC.	70% correct A 14% · B 9% · C 70% · D 7%
☐	6	B	Moving pH away from the optimum can denature enzymes. Cold only slows enzymes down (it doesn't denature them), and the two pathways use different enzymes.	78% correct A 10% · B 78% · C 6% · D 5%
☐	7	B	The virus spreads in bat body fluids, so cover horse feed and water. Vaccinating every wild bat isn't practical. <i>Most popular wrong answer: D (16%).</i>	71% correct A 6% · B 71% · C 6% · D 16%
☐	8	C	Communicating all results honestly is integrity.	71% correct A 13% · B 12% · C 71% · D 3%
☐	9	D	Same basic structure, different function, common ancestry: homologous structures.	74% correct A 2% · B 10% · C 14% · D 74%
☐	10	C	Rotating branches around a node doesn't change relationships, so both trees show the same relationships. No living group is "more advanced". <i>Most popular wrong answer: A (16%).</i>	71% correct A 16% · B 10% · C 71% · D 3%
☐	11	C	mtDNA is inherited only from the mother, without recombination, so it tracks maternal lineages over many generations.	71% correct A 8% · B 7% · C 71% · D 13%
☐	12	D	Every nucleotide has a pentose sugar. A thymine nucleotide (in DNA) has deoxyribose, and uracil nucleotides belong to RNA. <i>Most popular wrong answer: B (20%).</i>	71% correct A 6% · B 20% · C 3% · D 71%
☐	13	B	The rough ER (organelle S) folds and transports the new protein towards the Golgi. Check each label against its organelle. <i>Most popular wrong answer: A (15%).</i>	74% correct A 15% · B 74% · C 8% · D 3%
☐	14	B	One cell contains many proteins with different structures and functions (the proteome). Not every protein has multiple chains, and not all genes are expressed all the time.	75% correct A 4% · B 75% · C 12% · D 8%
☐	15	D	Systematic errors shift every reading from the true value by a consistent amount, which affects accuracy. Random errors cause spread (precision). <i>Most popular wrong answer: B (21%).</i>	71% correct A 5% · B 21% · C 3% · D 71%

✓	Q	Ans	Why	How Victoria went
☐	16	A	Stomata are openings in the leaf surface. The cuticle and hairs are barriers to entry.	72% correct A 72% · B 11% · C 12% · D 5%
☐	17	C	Electrophoresis separates fragments by size. Cutting is done by restriction enzymes, and amplifying by PCR. <i>Most popular wrong answer: B (17%).</i>	78% correct A 3% · B 17% · C 78% · D 2%
☐	18	D	The Calvin cycle uses CO ₂ , ATP and NADPH. NADH belongs to respiration.	74% correct A 6% · B 9% · C 11% · D 74%
☐	19	B	Lactate forms without oxygen, and the rate is highest at the warmest temperature that hasn't denatured the enzymes (20 °C here, from the data). <i>Most popular wrong answer: C (20%).</i>	77% correct A 2% · B 77% · C 20% · D 1%
☐	20	B	Inflammation is part of the innate immune response, so an anti-inflammatory dampens it.	70% correct A 13% · B 70% · C 9% · D 8%
☐	21	C	Only the information given supports a similar group size. Nothing tells you about random selection or anyone leaving the study. <i>Most popular wrong answer: A (17%).</i>	77% correct A 17% · B 3% · C 77% · D 3%
☐	22	B	Human constant regions are less likely to be recognised as foreign, so the patient is less likely to mount an immune response against the drug.	75% correct A 8% · B 75% · C 14% · D 2%
☐	23	A	Dendritic cells carry antigens to lymph nodes and present them to helper T cells. Lymph flows one way only. <i>Most popular wrong answer: B (16%).</i>	71% correct A 71% · B 16% · C 10% · D 3%
☐	24	A	The body makes its own antibodies and memory cells, so it is active, and it comes from a vaccine, so it is artificial. <i>Most popular wrong answer: D (15%).</i>	78% correct A 78% · B 3% · C 3% · D 15%
☐	25	B	Zika spreads by mosquito bites, not breathing, so face masks do little. Covering skin blocks bites. <i>Most popular wrong answer: D (16%).</i>	71% correct A 7% · B 71% · C 6% · D 16%
☐	26	D	All mammals have hair, so it can't separate them. Molecular sequences and bone structure vary between species.	73% correct A 10% · B 13% · C 4% · D 73%
☐	27	A	Go from biggest to smallest group: mammals ⊃ primates ⊃ hominoids ⊃ hominins.	73% correct A 73% · B 14% · C 6% · D 6%
☐	28	B	Humans have a very large brain for their body mass, so pick the point furthest above the average-mammal line. <i>Most popular wrong answer: D (16%).</i>	77% correct A 2% · B 77% · C 5% · D 16%
☐	29	D	A large carrying angle puts the knees under the centre of gravity, which suits bipedalism. Chimps have almost no carrying angle. <i>Most popular wrong answer: B (17%).</i>	72% correct A 2% · B 17% · C 9% · D 72%
☐	30	C	The ETC produces water, lots of ATP and NAD ⁺ . NADH donates electrons and H ⁺ , and oxygen is the final electron acceptor.	77% correct A 7% · B 7% · C 77% · D 10%
☐	31	A	Photolysis happens in the light-dependent stage on the thylakoid membranes (grana). <i>Most popular wrong answer: B (20%).</i>	77% correct A 77% · B 20% · C 1% · D 2%

✓	Q	Ans	Why	How Victoria went
☐	32	D	Cellulose-rich plant material can be broken down to glucose for fermentation. Microorganisms are required, CO ₂ is also produced, and each step uses a different enzyme. <i>Most popular wrong answer: C (15%).</i>	72% correct A 6% · B 8% · C 15% · D 72%
☐	33	C	Antibodies crossing the placenta give passive (natural) immunity. Injected antigens give active immunity, and antibiotics give no immunity at all. <i>Most popular wrong answer: B (17%).</i>	70% correct A 10% · B 17% · C 70% · D 3%
☐	34	C	A cell that recognises an antigen on an APC and then activates other lymphocytes is a helper T cell.	77% correct A 11% · B 5% · C 77% · D 6%
☐	35	D	Humans chose the breeders, so this is selective breeding, not natural selection.	78% correct A 7% · B 3% · C 12% · D 78%
☐	36	B	Yeast fermentation gives ethanol + CO ₂ , and it needs oxygen to be absent (anaerobic).	75% correct A 13% · B 75% · C 3% · D 8%
☐	37	D	PCR repeats denature (≈95 °C), anneal (≈55 °C) and extend (≈72 °C, Taq polymerase) many times. Primers attach during annealing, and RNA polymerase isn't involved.	72% correct A 12% · B 11% · C 5% · D 72%
☐	38	A	Lactic acid fermentation happens in animal cells (in the cytosol). Ethanol + CO ₂ is the yeast/plant pathway. <i>Most popular wrong answer: D (15%).</i>	70% correct A 70% · B 5% · C 10% · D 15%
☐	39	B	An opposable thumb is a primate trait. Tails and fur are shared with many mammals, and primates have forward-facing eyes. <i>Most popular wrong answer: A (16%).</i>	75% correct A 16% · B 75% · C 7% · D 2%
☐	40	C	A transitional fossil supports a common ancestor between the groups. It can't give absolute ages or show which animals were most common.	79% correct A 7% · B 9% · C 79% · D 6%

Multiple-choice score: _____ / 40

Section B – Short-answer questions

Question 1 (7 marks) · state average 4.1 / 7 (59%)

2022 VCAA exam · Section B · Question 4

a. Roles of organelles in exporting ovalbumin

- ☐ The rough ER (ribosomes) makes and transports the protein to the Golgi
- ☐ The Golgi modifies and packages it into vesicles
- ☐ Vesicles fuse with the plasma membrane, releasing the protein by exocytosis

Answer / notes: Ribosomes/RER, then Golgi (packaging), then vesicle, then exocytosis. Mitochondria supply ATP.

State average 1.7 / 3 (57%); 42% scored full marks. Mark distribution: 0: 29% · 1: 12% · 2: 17% · 3: 42%

Your mark: _____ / 3

b. Discuss an ethical concern, propose a solution and name the ethical concept

- ☐ An ethical concern identified (e.g. chicken welfare in cramped cages)
- ☐ Its cause and effect explained (stress, disease, injury)
- ☐ A feasible solution (e.g. free-range standards, or recombinant ovalbumin)
- ☐ An ethical concept/approach named and linked (e.g. non-maleficence, virtue/consequences-based)

Answer / notes: Concern, then cause and effect, then fix, then concept. For example, caged hens suffer (non-maleficence), so switch to recombinant ovalbumin.

State average 2.4 / 4 (60%); 21% scored full marks. Mark distribution: 0: 14% · 1: 12% · 2: 22% · 3: 32% · 4: 21%

Your mark: _____ / 4

Question 2 (7 marks) · state average 3.9 / 7 (56%)

2022 VCAA exam · Section B · Question 6

a. Compare the roles of natural killer cells and cytotoxic T cells

- ☐ Both destroy virus-infected cells (induce apoptosis)
- ☐ Both detect infected cells via MHC I markers (NK cells respond to altered or missing MHC I)
- ☐ NK cells are innate / non-specific / act early
- ☐ Cytotoxic T cells are adaptive / specific / act later after clonal expansion

Answer / notes: Similarities (both kill infected cells) plus differences (innate and non-specific vs adaptive and specific). Use comparative language.

State average 2.2 / 4 (55%); 30% scored full marks. Mark distribution: 0: 22% · 1: 20% · 2: 12% · 3: 16% · 4: 30%

Your mark: _____ / 4

b.i. What change in the virus alters the antigen

- ☐ Mutation (in the antigen genes)

Answer / notes: Mutation.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: _____ / 1

b.ii. Why flu vaccination is yearly, in terms of antigenic drift

- ☐ Drift changes the antigens, so existing memory cells/antibodies no longer match
- ☐ The new vaccine contains updated antigens, producing new specific memory cells

Answer / notes: Mutations change the antigens, so last year's memory cells don't recognise them, and the vaccine is updated each year.

State average 1.0 / 2 (50%); 33% scored full marks. Mark distribution: 0: 34% · 1: 34% · 2: 33%

Your mark: _____ / 2

Question 3 (11 marks) · state average 6.0 / 11 (55%)

2023 VCAA exam · Section B · Question 6

a. Outline how human insulin is made using recombinant plasmids

- ☐ The genes for chain A and chain B are isolated and inserted into separate plasmids
- ☐ The same restriction enzyme cuts the gene and the plasmid (matching sticky ends)
- ☐ DNA ligase joins the gene into the plasmid
- ☐ Bacteria are transformed (heat shock) and selected (antibiotic resistance/reporter gene)
- ☐ Bacteria express the chains, which are purified and joined by disulfide bridges

Answer / notes: Two genes into two plasmids, cut, ligate, transform, select, express, then purify and combine A + B.

State average 2.4 / 5 (48%); 18% scored full marks. Mark distribution: 0: 29% · 1: 9% · 2: 12% · 3: 12% · 4: 20% · 5: 18%

Your mark: _____ / 5

b. Which insulin is rapid-acting compared with regular, with graph data

- ☐ Lispro
- ☐ It peaks sooner (~2 h vs ~5 h) or higher
- ☐ Its activity lasts a shorter time (~5 h vs ~12 h)

Answer / notes: Lispro: it peaks at about 2 h (vs about 5 h) and is gone by about 5 h. Quote numbers from the graph.

State average 2.3 / 3 (77%); 36% scored full marks. Mark distribution: 0: 2% · 1: 4% · 2: 58% · 3: 36%

Your mark: _____ / 3

c.i. Hierarchical level of insulin

- ☐ Quaternary

Answer / notes: Quaternary (two polypeptide chains).

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: _____ / 1

c.ii. What to consider when changing insulin's primary structure, with justification

- ☐ The change must keep the overall 3D (tertiary/quaternary) shape and binding site
- ☐ So the insulin still binds its receptor and functions

Answer / notes: Don't disrupt the folding. The shape has to still fit the insulin receptor.

State average 0.6 / 2 (30%); 12% scored full marks. Mark distribution: 0: 56% · 1: 32% · 2: 12%

Your mark: _____ / 2

Question 4 (5 marks) · state average 2.8 / 5 (56%)

2023 VCAA exam · Section B · Question 9

a. Two factors in the flowering data that may have contributed to speciation

- ☐ The species flower at different times (different weeks)
- ☐ Peak flowering of males and females of one species doesn't overlap with the other species, so there is little cross-pollination

Answer / notes: Different flowering times mean temporal isolation, and peak male/female flowering doesn't overlap between species.

State average 0.9 / 2 (45%); 16% scored full marks. Mark distribution: 0: 30% · 1: 54% · 2: 16%

Your mark: _____ / 2

b.i. Ideal pH and altitude for each species from the graphs

- ☐ H. forsteriana: pH 8, 31–60 m; H. belmoreana: pH 6, > 121 m

Answer / notes: Forsteriana: pH 8, 31–60 m. Belmoreana: pH 6, above 121 m.

State average 0.9 / 1 (90%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

Your mark: _____ / 1

b.ii. How the pH and altitude differences could have led to speciation

- ☐ Different soils/altitudes impose different selection pressures, so different alleles are selected (e.g. forsteriana adapted to basic, low-altitude soils)
- ☐ This shifted flowering times, stopping gene flow (reproductive isolation), so the gene pools diverged

Answer / notes: Different soils select for different alleles, which shifted flowering times, which cut gene flow, so sympatric speciation.

State average 1.0 / 2 (50%); 31% scored full marks. Mark distribution: 0: 35% · 1: 34% · 2: 31%

Your mark: _____ / 2

Question 5 (7 marks) · state average 4.1 / 7 (59%)

2023 VCAA exam · Section B · Question 11

a. Two controlled factors

- ☐ One (e.g. 10 mL of medium)
- ☐ Another (e.g. 0.1 mL transferred every 24 h, temperature, starting glucose/citrate)

Answer / notes: For example, the medium volume and the 0.1 mL daily transfer.

State average 1.3 / 2 (65%); 46% scored full marks. Mark distribution: 0: 14% · 1: 40% · 2: 46%

Your mark: _____ / 2

b. Expected effect of the citrate-using mutation on later generations, and why

- ☐ The mutant allele increases in frequency (mutants reproduce more)
- ☐ Because they have an extra food/energy source (a selective advantage)

Answer / notes: Citrate users have an extra food source, reproduce more, and the allele spreads.

State average 1.4 / 2 (70%); 52% scored full marks. Mark distribution: 0: 18% · 1: 29% · 2: 52%

Your mark: _____ / 2

c. Design an experiment on glucose concentration and the frequency of the citrate phenotype

- ☐ Controlled variables stated (medium volume, time, temperature, citrate amount)
- ☐ A control group (e.g. no glucose)
- ☐ Repeats / large samples, and the measurement of mutant frequency

Answer / notes: Vary the glucose (e.g. 0, 0.5, 1, 2%), keep everything else constant, include a 0% control, replicate, and measure the proportion of citrate users.

State average 1.4 / 3 (47%); 17% scored full marks. Mark distribution: 0: 25% · 1: 30% · 2: 28% · 3: 17%

Your mark: _____ / 3

Question 6 (4 marks) · state average 2.2 / 4 (55%)

2024 VCAA exam · Section B · Question 6

a. Compare the variable regions of the two antibodies

- ☐ They bind different sites (epitopes) on HER2
- ☐ So their variable regions differ (each is complementary to its own site)

Answer / notes: They bind different parts of HER2, so their variable regions have different shapes. Mention both antibodies.

State average 1.2 / 2 (60%); 48% scored full marks. Mark distribution: 0: 28% · 1: 24% · 2: 48%

Your mark: _____ / 2

b. Two ways the mAbs could help treat the cancer

- ☐ Block the receptor, so growth signals are stopped and division slows
- ☐ Deliver drugs/radioisotopes specifically to cancer cells, or flag them for immune destruction

Answer / notes: They block the HER2 growth signal, and they can carry toxins or attract immune cells specifically to cancer cells.

State average 1.0 / 2 (50%); 32% scored full marks. Mark distribution: 0: 33% · 1: 35% · 2: 32%

Your mark: _____ / 2

Question 7 (10 marks) · state average 5.8 / 10 (58%)

2024 VCAA exam · Section B · Question 7

a. An action that may explain the fall in cases from 1994, and how

- ☐ Vaccination (or isolation/education)
- ☐ Increases herd immunity / reduces available hosts
- ☐ So transmission falls and there are fewer reported cases

Answer / notes: Vaccination campaigns built herd immunity, which cut transmission, so cases fell.

State average 2.5 / 3 (83%); 61% scored full marks. Mark distribution: 0: 3% · 1: 10% · 2: 26% · 3: 61%

Your mark: _____ / 3

b. A likely source of the new cases and two benefits of listing locations

- ☐ Source: an infected traveller returning from overseas (or an unvaccinated contact)
- ☐ Benefit 1 (e.g. exposed people can monitor symptoms/get tested)
- ☐ Benefit 2 (e.g. people can isolate or avoid those places)

Answer / notes: Most likely an infected traveller. Listing sites lets exposed people watch for symptoms and isolate.

State average 1.9 / 3 (63%); 27% scored full marks. Mark distribution: 0: 6% · 1: 23% · 2: 43% · 3: 27%

Your mark: _____ / 3

c. Describe two factors driving the emergence of new pathogens

- ☐ Factor 1 named (e.g. zoonosis / closer contact with wildlife)
- ☐ Description of how it produces new human pathogens
- ☐ Factor 2 named (e.g. antigenic shift, climate change, travel into new areas)
- ☐ Description of how it produces new human pathogens

Answer / notes: For example, zoonosis (animal viruses jumping to humans as habitats are cleared) and antigenic shift (viruses reassorting into a novel strain). The state averaged 1.4/4.

State average 1.4 / 4 (35%); 7% scored full marks. Mark distribution: 0: 33% · 1: 21% · 2: 33% · 3: 6% · 4: 7%

Your mark: _____ / 4

Question 8 (5 marks) · state average 2.8 / 5 (56%)

2025 VCAA exam · Section B · Question 1

a. Draw the processed mRNA and label three modifications

- ☐ Introns removed / exons spliced together
- ☐ Methyl-G (modified guanine) cap at the 5' end
- ☐ Poly-A tail at the 3' end (or alternative splicing)

Answer / notes: Exons 1–2–3 joined with a 5' cap and a 3' poly-A tail.

State average 2.0 / 3 (67%); 56% scored full marks. Mark distribution: 0: 25% · 1: 5% · 2: 14% · 3: 56%

Your mark: _____ / 3

b. Complete steps 4 and 8 of recombinant insulin production

- ☐ Step 4: DNA ligase joins each insulin gene (A or B) into its own plasmid (forms phosphodiester bonds)
- ☐ Step 8: chains A and B are combined (disulfide bonds) to form functional insulin

Answer / notes: Ligase seals chain A's gene into one plasmid and chain B's into another. Later, chains A and B are joined to make insulin.

State average 0.8 / 2 (39%); 24% scored full marks. Mark distribution: 0: 46% · 1: 30% · 2: 24%

Your mark: _____ / 2

Question 9 (10 marks) · state average 5.7 / 10 (57%)

2025 VCAA exam · Section B · Question 6

a. Type of immunity the babies get, justified

- ☐ Natural passive immunity
- ☐ Passive because the baby receives ready-made antibodies (no memory cells); natural because they come from the mother (placenta/breastmilk)

Answer / notes: Natural passive: maternal antibodies cross the placenta. The mother's immunity is artificial active, but the baby's isn't.

State average 1.1 / 2 (54%); 40% scored full marks. Mark distribution: 0: 32% · 1: 27% · 2: 40%

Your mark: _____ / 2

b. Why a booster is advised in a later pregnancy

- ☐ Antibody and memory-cell levels decline over time
- ☐ A booster raises antibody levels so enough pass to the new baby

Answer / notes: Antibodies wane, so a booster makes sure the next baby gets enough. Memory cells don't cross the placenta.

State average 1.1 / 2 (52%); 34% scored full marks. Mark distribution: 0: 29% · 1: 37% · 2: 34%

Your mark: _____ / 2

c. One economic benefit of vaccinating pregnant women

- ☐ A stakeholder saves money (e.g. fewer infant hospitalisations)
- ☐ Explanation: vaccinating is cheaper than treating severe RSV (or parents lose less work time)

Answer / notes: Fewer infant hospital admissions, and prevention costs less than intensive care.

State average 1.5 / 2 (74%); 65% scored full marks. Mark distribution: 0: 17% · 1: 18% · 2: 65%

Your mark: _____ / 2

d. How monoclonal antibodies reduce RSV severity in infants

- ☐ They bind specifically to RSV antigens
- ☐ This neutralises or agglutinates the virus or tags it for phagocytes, limiting spread

Answer / notes: The mAbs bind RSV antigens and neutralise the virus or mark it for destruction, so less infection. More than half the state scored zero, so name the mechanism.

State average 0.7 / 2 (36%); 26% scored full marks. Mark distribution: 0: 54% · 1: 20% · 2: 26%

Your mark: _____ / 2

e. Another way to reduce RSV spread to infants, justified

- ☐ A valid measure (masks, hand hygiene, isolate sick people, avoid crowds)
- ☐ Justification linked to droplet transmission

Answer / notes: Visitors wear masks or stay away when sick, which reduces the respiratory droplets reaching the infant.

State average 1.4 / 2 (70%); 61% scored full marks. Mark distribution: 0: 20% · 1: 18% · 2: 61%

Your mark: _____ / 2

Question 10 (14 marks) · state average 8.0 / 14 (57%)

2025 VCAA exam · Section B · Question 8

a. Two reasons tadpoles were chosen

- ☐ A valid reason from the article (e.g. laid in thousands, so a large sample)
- ☐ A second (e.g. transparent skin lets light reach the cyanobacteria)

Answer / notes: Large numbers (big samples) and transparent skin (light gets in).

State average 1.5 / 2 (74%); 55% scored full marks. Mark distribution: 0: 6% · 1: 40% · 2: 55%

Your mark: _____ / 2

b.i. Two molecules other than enzymes that cyanobacteria need to produce oxygen

- ☐ Water
- ☐ Chlorophyll (or NADP⁺/ADP + P_i)

Answer / notes: Water and chlorophyll. CO₂ and glucose are light-independent, and light isn't a molecule.

State average 0.9 / 2 (44%); 18% scored full marks. Mark distribution: 0: 28% · 1: 55% · 2: 18%

Your mark: _____ / 2

b.ii. Role of each molecule in producing oxygen

- ☐ Water is split (photolysis), releasing O₂ and H⁺/electrons
- ☐ Chlorophyll absorbs light energy to drive the splitting of water

Answer / notes: Water is split to give O₂. Chlorophyll captures the light energy that drives it.

State average 0.9 / 2 (46%); 28% scored full marks. Mark distribution: 0: 35% · 1: 37% · 2: 28%

Your mark: _____ / 2

c. Why brain activity stopped under long hypoxia

- ☐ Without oxygen, aerobic respiration (the ETC) can't occur
- ☐ Anaerobic respiration gives far less ATP, not enough for neurons

Answer / notes: No O₂ means no aerobic respiration, and the little ATP left isn't enough to keep neurons firing.

State average 1.0 / 2 (48%); 20% scored full marks. Mark distribution: 0: 23% · 1: 58% · 2: 20%

Your mark: _____ / 2

d. Why brain activity took time to restart, using a biochemical pathway

- ☐ Photosynthesis first has to produce oxygen after the light comes on
- ☐ Then aerobic respiration must use it to generate enough ATP for neuron activity

Answer / notes: Light, then photosynthesis makes O₂, then respiration makes ATP, then neurons restart. Each step takes time.

State average 0.9 / 2 (46%); 31% scored full marks. Mark distribution: 0: 38% · 1: 30% · 2: 31%

Your mark: _____ / 2

e. Two events along the electron transport chain

- ☐ NADH/FADH₂ unload electrons and H⁺
- ☐ Oxygen accepts electrons and forms water, and 26–28 ATP are produced

Answer / notes: Loaded coenzymes drop off electrons, O₂ is the final acceptor (making water), and most ATP is made here.

State average 1.2 / 2 (61%); 50% scored full marks. Mark distribution: 0: 27% · 1: 23% · 2: 50%

Your mark: _____ / 2

f. Which student's ethical view (beneficence vs non-maleficence) you agree with, justified from the article

- ☐ Chooses a student and correctly defines the relevant principle
- ☐ Justifies using the article (e.g. potential to save human lives outweighs harm; brain activity was restarted)

Answer / notes: For example, Student X: beneficence is about maximising benefit, and this could save lives in emergencies while the tadpoles recovered.

State average 1.6 / 2 (80%); 69% scored full marks. Mark distribution: 0: 8% · 1: 23% · 2: 69%

Your mark: _____ / 2

Your result

Total: _____ / 120 State average on these questions: **75 / 120 (62%)**

Topics to revise:

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